A PROJECT REPORT

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submitted to

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CNN-Based Project Assessment

Case Study B: Plant Disease Detection

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Abstract

Detecting plant diseases early is essential for healthy crop yields, but doing it manually is slow and prone to errors. This project automates the detection of apple leaf diseases using Deep Learning and Computer Vision. Using the Plant Pathology 2020 dataset, the goal was to classify images into four categories: Healthy, Rust, Scab, and Multiple Diseases. The biggest challenge was a severe data imbalance—there were many 'Rust' images but very few 'Multiple Diseases' images. To solve this, the data was balanced using stratified splitting, image augmentation, and mathematical class weights. The project tested three different models: a basic custom CNN, an augmented custom CNN, and a pre-trained ResNet50 model. The results showed that basic models struggled with complex leaf textures, achieving only 44.89% and 26.28% accuracy. However, by using Transfer Learning with the ResNet50 model, the system achieved a final test accuracy of 85.04%. This project proves that deep, pre-trained networks combined with proper data balancing are highly effective for solving real-world agricultural vision problems.

1 Introduction

This report details the development, training, and comparative evaluation of three Convolutional Neural Network (CNN) architectures for the automated detection of apple foliar diseases. The project utilizes the Plant Pathology 2020 FGVC7 dataset[1], which consists of high-resolution RGB images captured in real orchard conditions. The objective is a multi-class classification task to categorize leaves into four distinct classes: Healthy, Scab, Rust, and Multiple Diseases. Automating this detection process holds significant real-world relevance for agricultural yield optimization and early intervention in crop pathology.

2 Dataset Description and Preprocessing Pipeline

2.1 Dataset Description

The dataset comprises 1,821 expert-annotated images originally formatted at high resolutions (e.g., 2048×1365 pixels). An initial class distribution analysis revealed a severe imbalance: Rust (622 samples) and Scab (592 samples) heavily dominate the dataset, while Healthy (516 samples) and Multiple Diseases (91 samples) are less frequent. This imbalance necessitates specific mitigation strategies to prevent the network from developing a bias toward the majority classes.[2]

2.2 Preprocessing Pipeline

To ensure robust model training and prevent data leakage, the following preprocessing steps were systematically applied:

- **Stratified Splitting:** Prior to any mathematical transformations, the dataset was partitioned into Training (70%), Validation (15%), and Test (15%) sets. A stratified split was enforced based on the target labels to guarantee that the critically underrepresented 'multiple_diseases' class maintained proportional representation across all subsets.
- **Class Weights Computation:** To further mitigate the class imbalance, balanced class weights were calculated exclusively on the training set. These weights explicitly penalize the loss function more heavily for misclassifying minority class samples during training.
- **Image Resizing:** All images were downscaled to 224×224 pixels. While this alters the original aspect ratio, it is a strict requirement to maintain tensor compatibility for the transfer learning architecture (ResNet50) utilized in Model 3.
- **Pixel Normalization & Data Typing:** For the custom CNNs (Models 1 and 2), pixel intensities were rescaled to the $[0, 1]$ range (divided by 255) and converted to `float32` tensors. This stabilizes gradient-based optimization by preventing large input variances from causing erratic weight updates.

3 Model Architectures

3.1 Model 1: Custom CNN (Baseline)

The baseline model features three sequential convolutional blocks. Each block utilizes 3×3 kernels with increasing filter depths (32, 64, 128), ReLU activations to introduce non-linearity, Batch Normalization to stabilize intermediate representations, and MaxPooling (2×2) to progressively down-sample spatial dimensions. The feature maps are flattened and passed to a 128-neuron Dense layer, followed by a Dropout layer (rate = 0.5) for regularization, and a 4-neuron Softmax classification head.

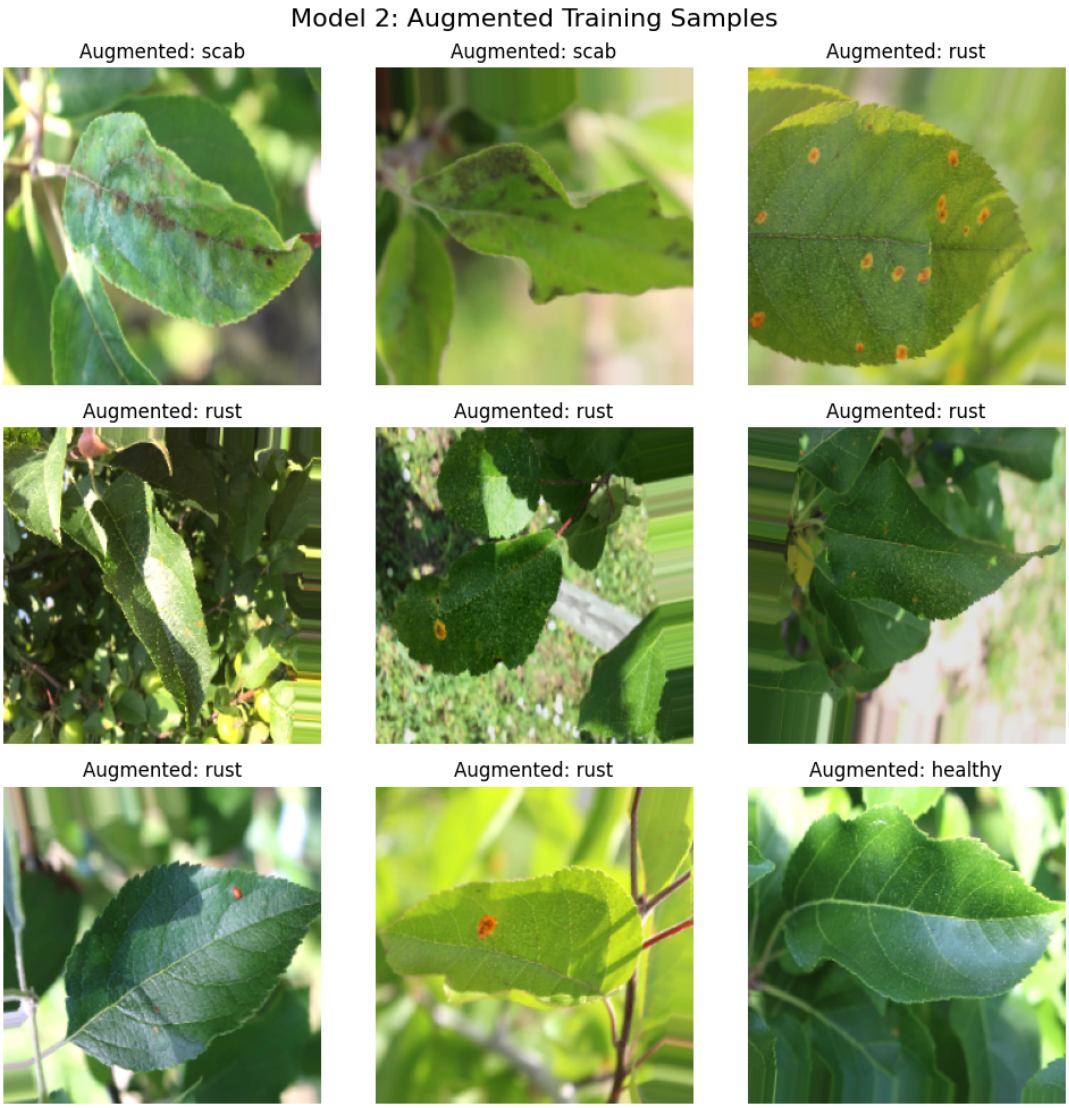

Figure 1: Grid of dynamically augmented training samples for Model 2.

3.2 Model 2: Custom CNN (Augmented)

Model 2 utilizes the exact architectural topology as Model 1. However, the training data is dynamically manipulated using an `ImageDataGenerator`. Spatial augmentations applied include random rotations ($\pm 15°$), width/height shifts (10%), zoom variance (20%), and horizontal/vertical flips. The validation and test sets remained strictly unaugmented to ensure a real-world evaluation baseline.

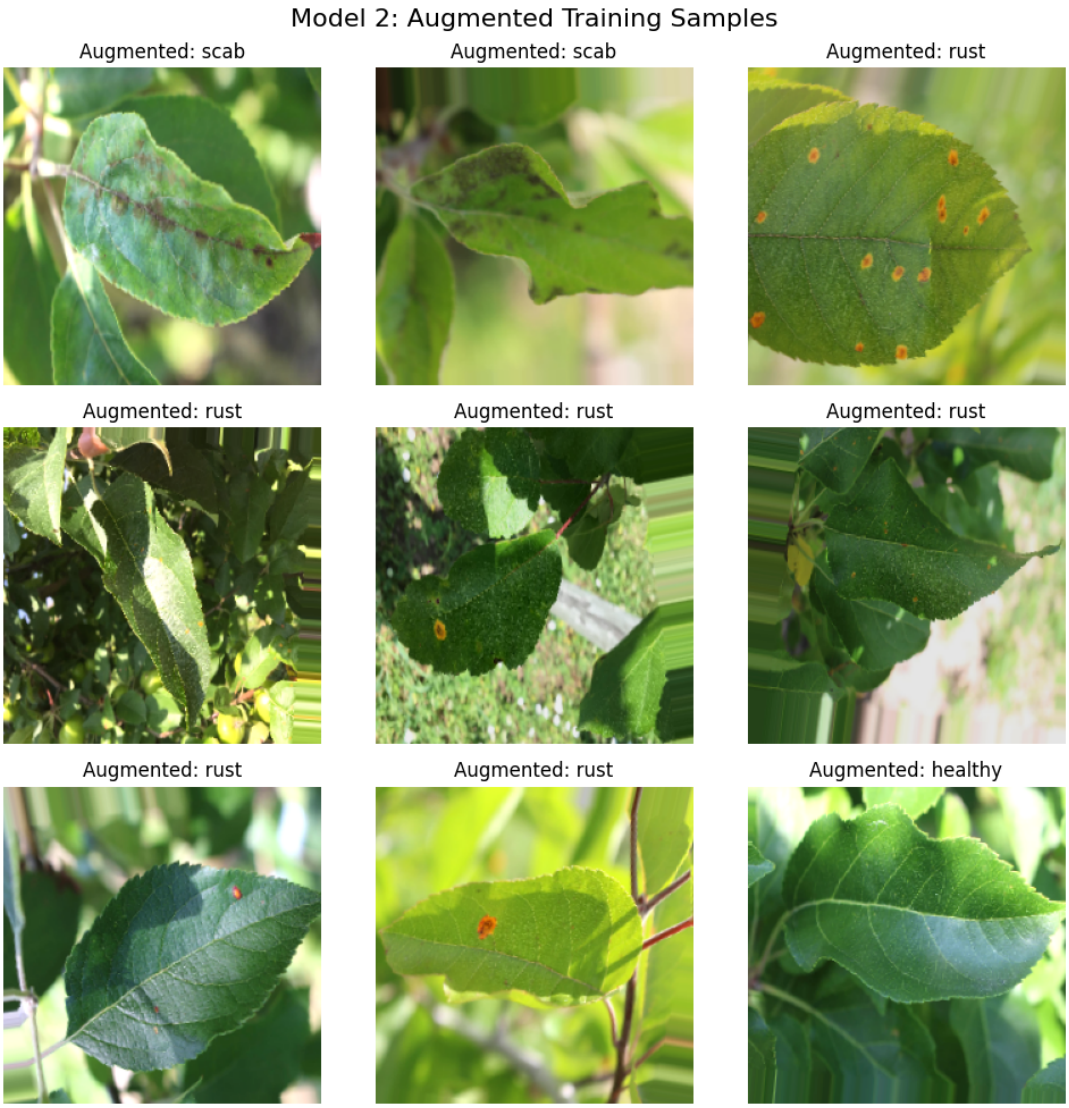

Figure 2: Grid of dynamically augmented training samples for Model 2.

3.3 Model 3: Transfer Learning (ResNet50)

Model 3 leverages the ResNet50 architecture, pre-trained on the ImageNet database. The original classification head was removed (`include_top=False`). The input pipeline was modified to replace $[0, 1]$ rescaling with ImageNet channel-wise mean subtraction. A custom head consisting of GlobalAveragePooling2D, a Dense layer (256 neurons, ReLU), Dropout (0.5), and a 4-neuron Softmax output was attached.

4 Training Configuration

All models were optimized using the Adam optimizer with an initial learning rate of $1e-3$ (reduced to $1e-5$ during Model 3 fine-tuning). The Categorical Cross-Entropy loss function was utilized, supplemented by the computed class weights.

To ensure computational efficiency and prevent overfitting, three callbacks were implemented:

1. **EarlyStopping:** Monitored validation loss with a patience of 4-5 epochs, restoring the best weights upon termination.

2. **ReduceLROnPlateau:** Reduced the learning rate by a factor of 0.2 if validation loss stagnated for 3 epochs.

3. **ModelCheckpoint:** Saved the optimal model weights based on minimum validation loss.

5 Results and Evaluation

The models were strictly evaluated against the 15% held-out test set.

Table 1: Summary Comparison Table of All Models

Model	**Test Accuracy**	**Weighted Precision**	**Weighted Recall**	**Weighted F1-Score**
Model 1 (Baseline)	44.89%	0.4422	0.4489	0.3998
Model 2 (Augmented)	26.28%	0.2196	0.2627	0.2124
Model 3 (ResNet50)	85.04%	0.8631	0.8503	0.8547

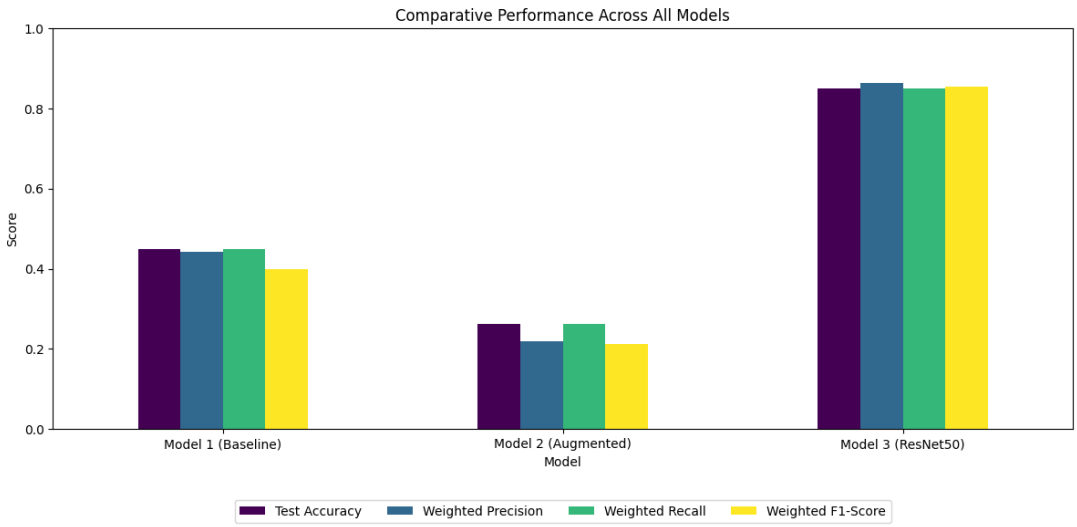

Figure 3: Grouped bar chart comparing performance metrics across all three models.

5.1 Model 1 Evaluation

The baseline CNN struggled to extract robust, generalized features, achieving 44.89% accuracy. The model demonstrated a heavy bias toward the dominant 'rust' and 'healthy' classes, confirming that raw class weights alone are insufficient for highly complex foliar textures without spatial augmentation.

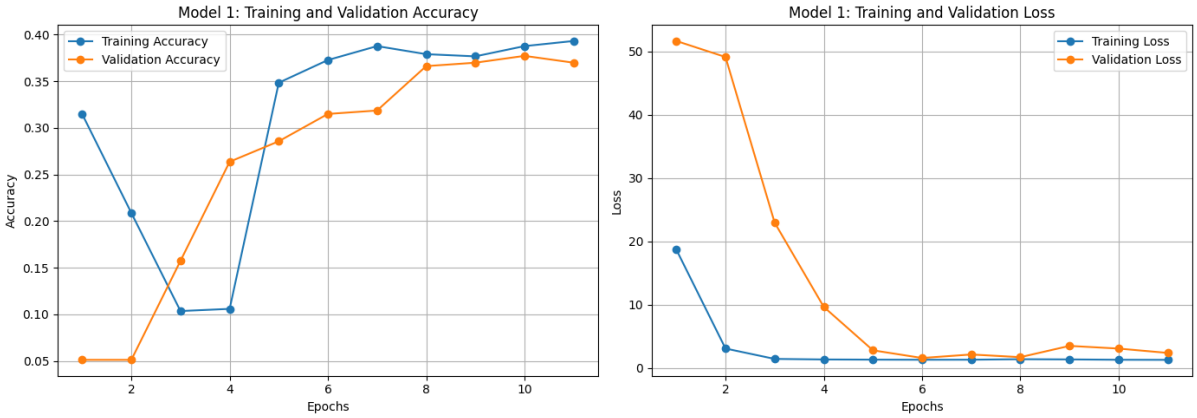

Figure 4: Training and Validation Accuracy and Loss for Model 1

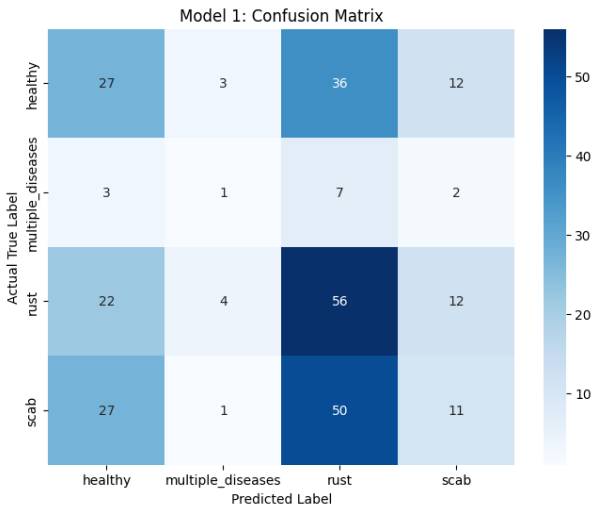

Figure 5: Confusion Matrix for Model 1 showing underfitting bias.

5.2 Model 2 Evaluation

While augmentation forced the network to recognize the minority 'multiple_diseases' class (recall improved from 8% to 31%), overall accuracy collapsed to 26.28%. The confusion matrix demonstrates catastrophic forgetting of the 'rust' class.

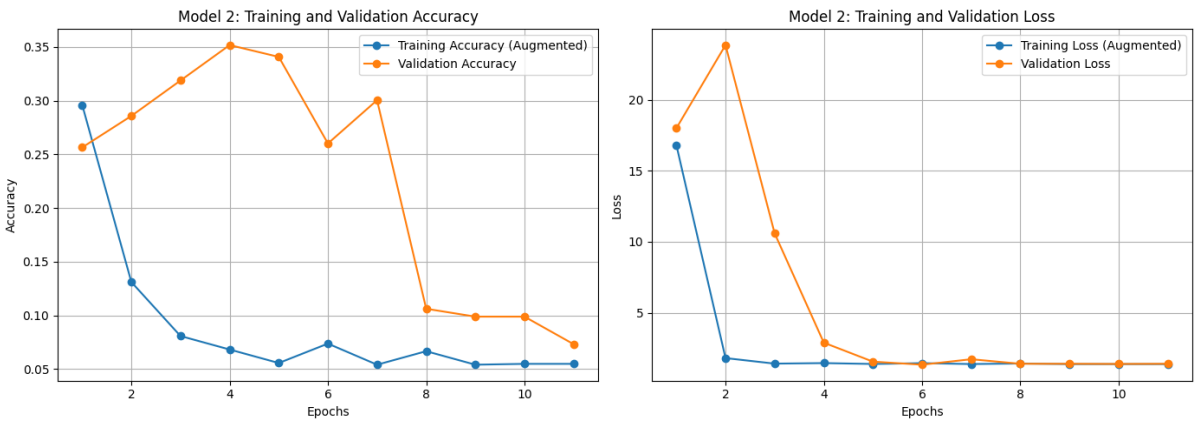

Figure 6: Training and Validation Accuracy and Loss for Model 2

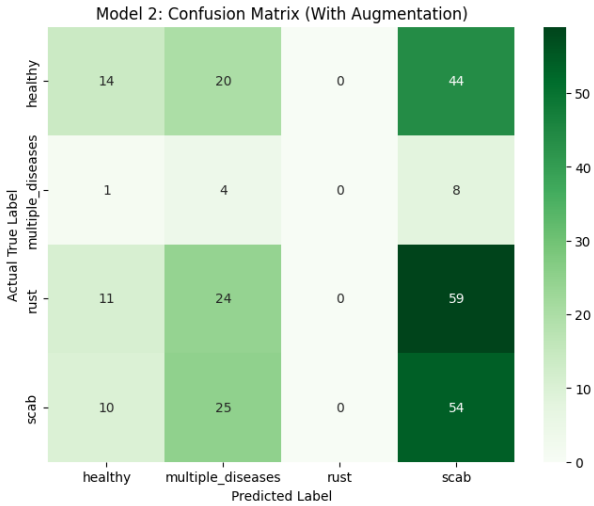

Figure 7: Confusion Matrix for Model 2 showing underfitting bias.

5.3 Model 3 Evaluation

Model 3 achieved an exceptional 85.04% accuracy. By applying a two-phase training approach—first freezing the base to train the dense head, then unfreezing the top 20 layers with a highly reduced learning rate—the model successfully adapted its deep feature hierarchies to the agricultural dataset.

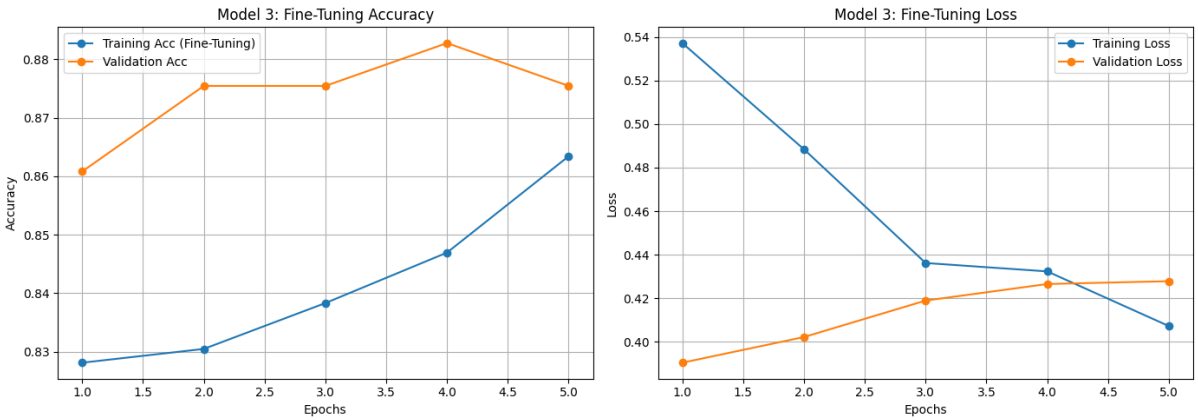

Figure 8: Training and Validation Accuracy and Loss for Model 3

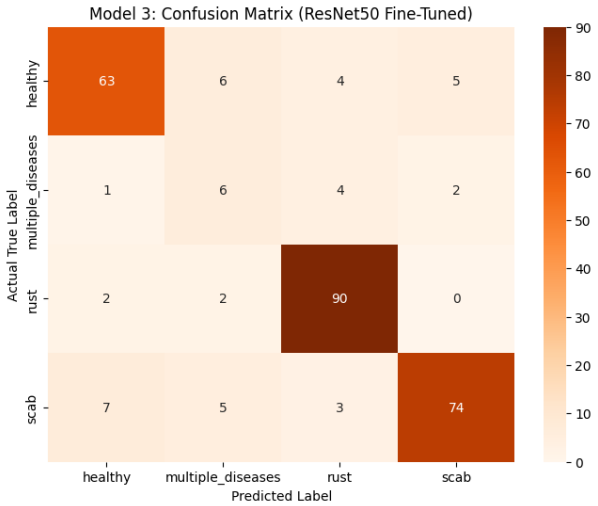

Figure 9: Confusion Matrix for the Fine-Tuned ResNet50 model.

6 Discussion and Justification

The comparative analysis demonstrates the profound impact of network depth and transfer learning on complex computer vision tasks. Model 1 and Model 2 lacked the structural parameter capacity to map high-variance foliar textures. Model 2's drop in accuracy explicitly highlights a fundamental deep learning concept: aggressive image augmentation acts as a strict regularizer, and if the underlying architecture lacks sufficient depth to process the expanded visual variance, the network will severely underfit. Model 3 (ResNet50) succeeded precisely because it leveraged millions of pre-trained parameters, overcoming the structural limitations that hindered the custom baseline CNNs.

6.1 Misclassification Analysis

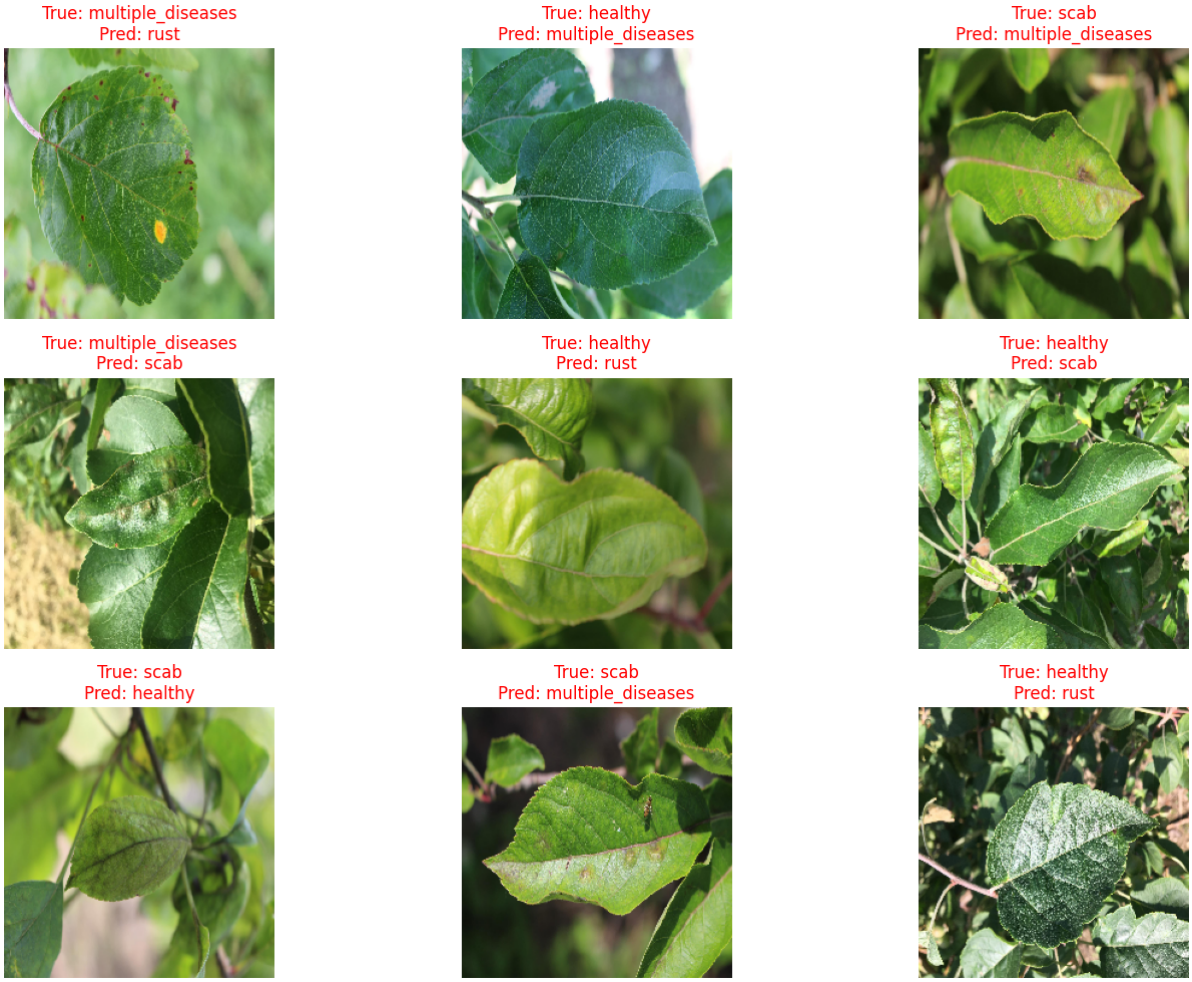

Figure 10: Sample of misclassified images by the optimal ResNet50 model.

Analyzing the specific failures of ResNet50 reveals logical error patterns:

- **The Multiple Diseases Ambiguity:** The model frequently confuses 'multiple_diseases' with singular 'rust' or 'scab' labels. Biologically, a leaf with multiple diseases is a visual amalgamation of both lesions; when one lesion type visually dominates the frame, the network confidently misclassifies it as a singular disease.
- **Physical Damage vs. Pathology:** The model occasionally categorizes physically damaged or heavily shadowed 'healthy' leaves as 'scab', incorrectly interpreting non-pathological dark spots as disease features.

7 Challenges Faced and Overcome

During the development of this deep learning pipeline, several technical and data-related challenges were encountered:

- **Severe Class Imbalance:** The dataset was highly skewed, with the majority of images belonging to the 'Rust' and 'Scab' classes, and very few representing 'Multiple Diseases' (only 91 images). This initially caused the baseline model to completely ignore the minority class. This was overcome by calculating and applying balanced class weights during training and ensuring proportional representation using stratified train-test splits.
- **Model Underfitting due to Augmentation:** When spatial data augmentation (rotations, flips, shifts) was introduced to Model 2 to prevent overfitting, the custom CNN's accuracy actually dropped. This revealed a structural challenge: the shallow custom network lacked the required parameter capacity (depth) to learn from heavily distorted images. This challenge directly justified the necessary pivot to a deep Transfer Learning architecture.
- **Visual Ambiguity of Pathologies:** Leaves classified as 'Multiple Diseases' inherently share visual characteristics with both Rust and Scab. The network struggled when one disease type visually dominated the image. Furthermore, physical leaf damage (like holes or dark shadows) was occasionally misinterpreted by the model as Scab lesions due to similar pixel intensities.
- **Transfer Learning Weight Destruction:** When initially attaching a randomly initialized dense classification head to the pre-trained ResNet50 base, large error gradients risked destroying the valuable pre-trained ImageNet weights. This was solved by utilizing a two-phase training approach: strictly freezing the base layers while the custom head stabilized, followed by a controlled unfreezing of the top 20 layers at a severely reduced learning rate ($1e-5$).

8 Conclusion

This project systematically evaluated three CNN architectures for plant disease detection. While shallow custom networks proved insufficient due to dataset complexity and class imbalance, the fine-tuned ResNet50 model demonstrated high commercial viability with an 85.04% test accuracy. The results definitively justify the implementation of transfer learning for high-resolution agricultural image classification. Future iterations should focus on expanding the minority class dataset and employing visual masking techniques to help the model differentiate between pathological lesions and standard physical leaf damage.

References

[1] `Kaggle competition` kaggle.com/c/plant-pathology-2020-fgvc7.

[2] `Research paper` arxiv.org/abs/2004.11958.